



L2-7 Chapter 2

Is a Laboratory Module the Starting Point of Organization, or Its Outcome?

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1. What Must Design Actually Organize?

Chapter 1 demonstrated that modularity is not a predefined design objective, but an organizational outcome that gradually emerges as laboratories continuously adapt to future uncertainty. As research directions, technologies, equipment, and operational requirements evolve, laboratories become increasingly difficult to organize around any predetermined state. Consequently, modularity should be understood not as an inherent characteristic of laboratory design, but as the result of long-term organizational evolution.

If this conclusion holds, one of the most fundamental assumptions behind laboratory modularity must be reconsidered.

In most discussions, a **Module** is treated as a predefined organizational unit. It may refer to a room, a collection of equipment, a furniture system, a workflow, or a functional area. Although different projects adopt different criteria for modularization, they all share the same underlying assumption:
Modules exist before organization, and modularity is achieved by organizing these predefined modules.

However, Chapter 1 established the opposite. If modularity is itself an emergent organizational outcome, then a Module cannot simultaneously serve as the starting point of organization. Otherwise, the existence of the Module is assumed before modularity is explained, resulting in circular reasoning. More importantly, such an assumption fails to answer a more fundamental question:

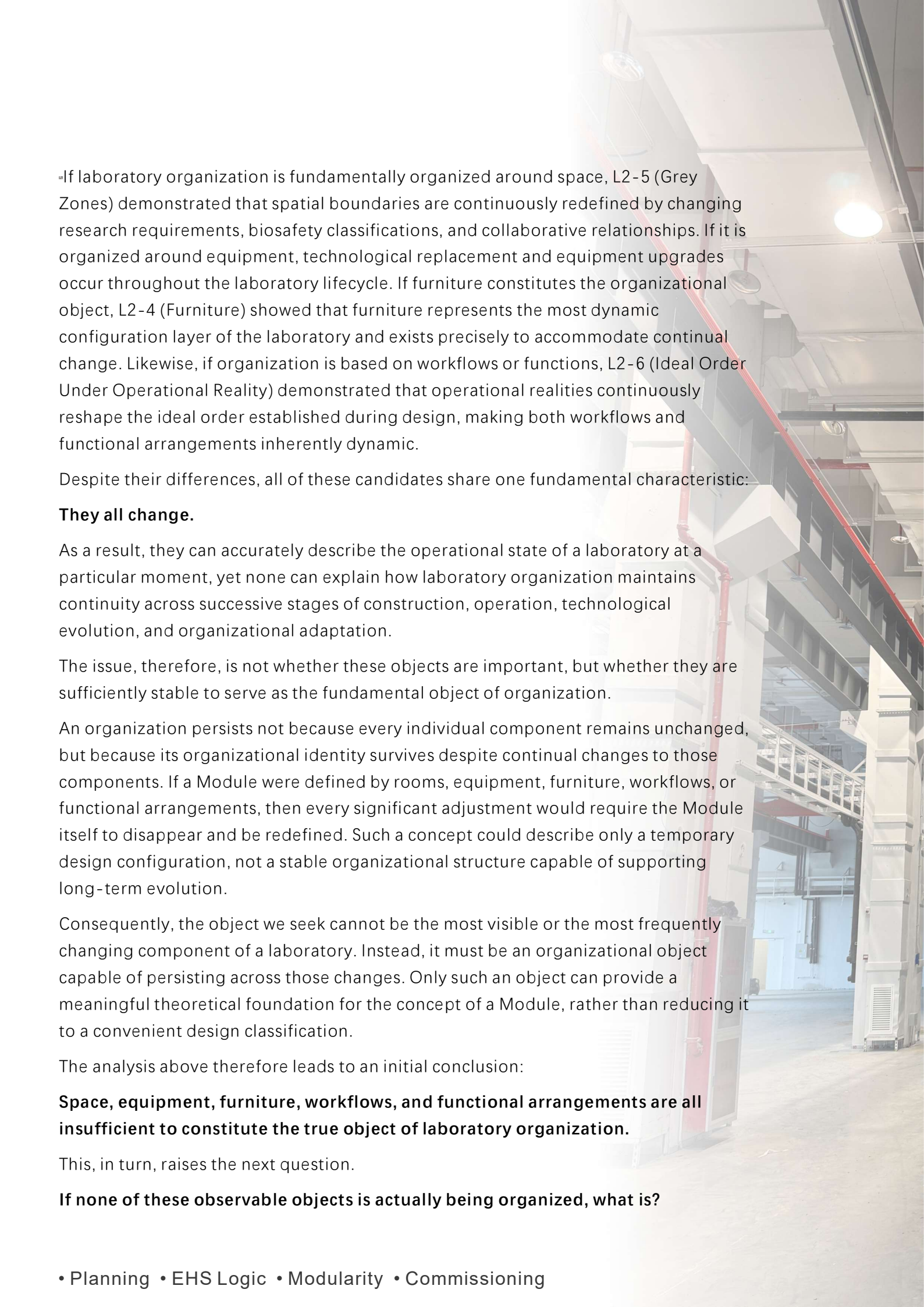
Why should these predefined modules remain valid as laboratories continuously evolve?

For this reason, this chapter does not begin by defining what a Module is. Instead, it first asks a more fundamental question:

What does laboratory design actually organize?

Although this appears to be a simple question of organizational focus, it determines whether a coherent theory of laboratory modularity can exist at all. Every organizational system must organize something. If Modules are organizational units, they must ultimately be derived from whatever that underlying object is. Without identifying that object, any discussion of Modules remains an empirical classification rather than a theoretical explanation of how laboratory organization evolves.

The most intuitive answers are also the most visible ones. Yet each has already been shown to be insufficient.



-If laboratory organization is fundamentally organized around space, L2-5 (Grey Zones) demonstrated that spatial boundaries are continuously redefined by changing research requirements, biosafety classifications, and collaborative relationships. If it is organized around equipment, technological replacement and equipment upgrades occur throughout the laboratory lifecycle. If furniture constitutes the organizational object, L2-4 (Furniture) showed that furniture represents the most dynamic configuration layer of the laboratory and exists precisely to accommodate continual change. Likewise, if organization is based on workflows or functions, L2-6 (Ideal Order Under Operational Reality) demonstrated that operational realities continuously reshape the ideal order established during design, making both workflows and functional arrangements inherently dynamic.

Despite their differences, all of these candidates share one fundamental characteristic:

They all change.

As a result, they can accurately describe the operational state of a laboratory at a particular moment, yet none can explain how laboratory organization maintains continuity across successive stages of construction, operation, technological evolution, and organizational adaptation.

The issue, therefore, is not whether these objects are important, but whether they are sufficiently stable to serve as the fundamental object of organization.

An organization persists not because every individual component remains unchanged, but because its organizational identity survives despite continual changes to those components. If a Module were defined by rooms, equipment, furniture, workflows, or functional arrangements, then every significant adjustment would require the Module itself to disappear and be redefined. Such a concept could describe only a temporary design configuration, not a stable organizational structure capable of supporting long-term evolution.

Consequently, the object we seek cannot be the most visible or the most frequently changing component of a laboratory. Instead, it must be an organizational object capable of persisting across those changes. Only such an object can provide a meaningful theoretical foundation for the concept of a Module, rather than reducing it to a convenient design classification.

The analysis above therefore leads to an initial conclusion:

Space, equipment, furniture, workflows, and functional arrangements are all insufficient to constitute the true object of laboratory organization.

This, in turn, raises the next question.

If none of these observable objects is actually being organized, what is?

2. What, Then, Constitutes a Module?

The previous section demonstrated that space, equipment, furniture, workflows, and functional arrangements cannot constitute the true object of laboratory organization. Although these are the most visible elements of a laboratory, they are all subject to continual change as research objectives, technologies, and operational requirements evolve. Consequently, they describe only the operational state of a laboratory at a particular moment, rather than explaining how laboratory organization maintains continuity across time.

If none of these observable objects can serve as the foundation of organization, then the true organizational object must possess a fundamentally different characteristic. It cannot depend on any particular room, piece of equipment, workflow, or physical configuration. Instead, it must remain meaningful despite changes in how those elements are implemented throughout the laboratory lifecycle.

Examining laboratory evolution from this perspective reveals a consistent pattern.

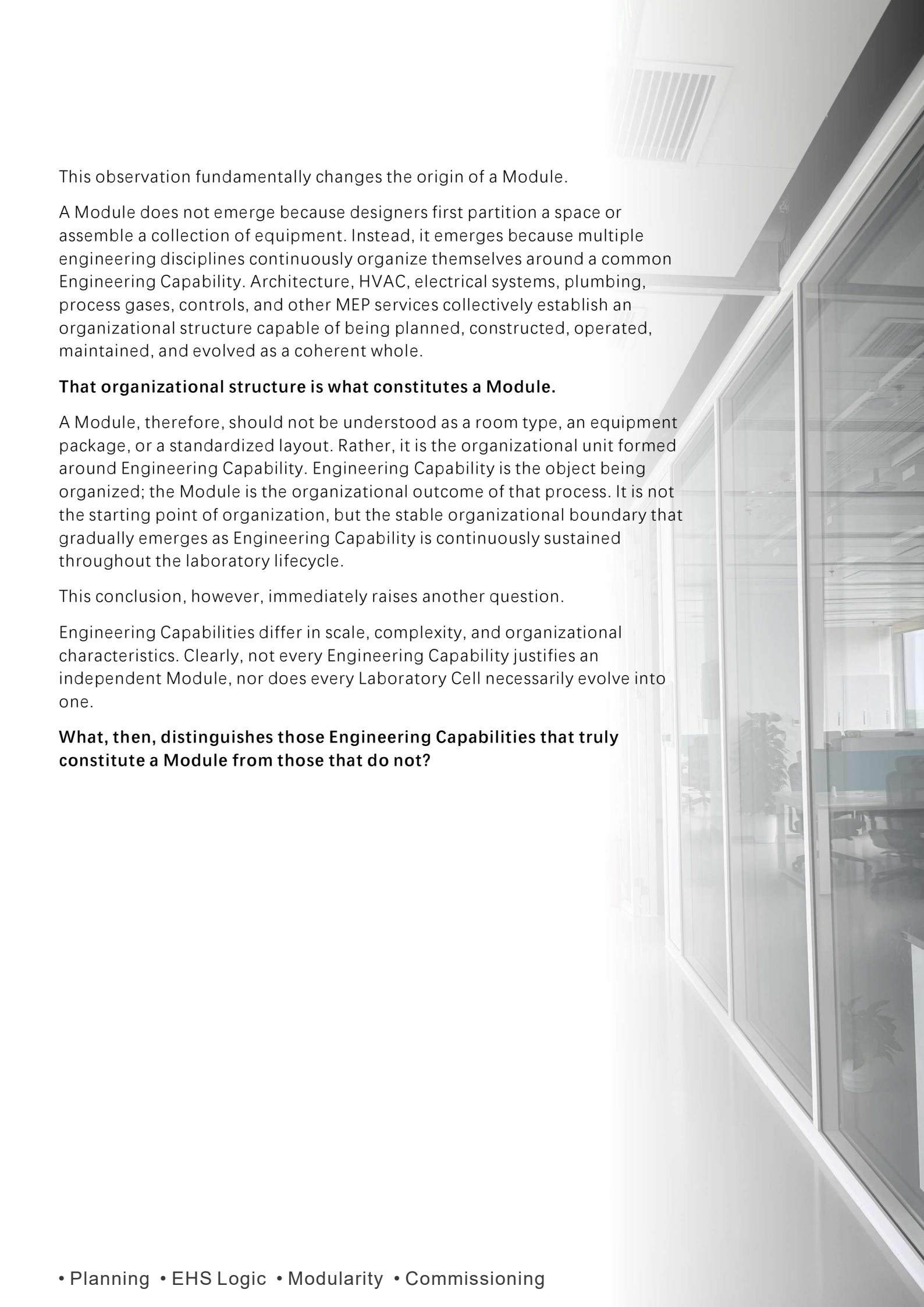
When experimental requirements change, rooms are reconfigured, equipment is replaced, furniture is rearranged, and workflows are redesigned. Yet these are not the objects that engineers actually reorganize. What must be continuously reorganized are the engineering systems that support laboratory operation. HVAC, electrical distribution, plumbing, process gases, control systems, and other MEP services are repeatedly reconfigured to satisfy new operational requirements. The physical resources change, but what is continuously organized is the engineering capability delivered by those resources.

This study defines such an organizational object as Engineering Capability: an engineering capability that can be continuously organized, maintained, and sustained independently of its specific physical implementation.

Importantly, this concept is not merely theoretical. In engineering practice, it is commonly manifested as a Laboratory Cell.

A Laboratory Cell, as defined in this study, is neither an architectural room nor a furniture layout. Rather, it is the smallest engineering organizational unit formed around a coherent Engineering Capability. Its boundary is not determined by walls, structural grids, or furniture arrangements, but by the engineering capability it supports together with the interfaces through which that capability is delivered.

Consequently, when equipment is upgraded, layouts are modified, or experimental procedures evolve, engineers typically reorganize the engineering resources within the Cell rather than redefining the Cell itself. What persists is therefore not a physical space, but the Engineering Capability carried by that Cell.



This observation fundamentally changes the origin of a Module.

A Module does not emerge because designers first partition a space or assemble a collection of equipment. Instead, it emerges because multiple engineering disciplines continuously organize themselves around a common Engineering Capability. Architecture, HVAC, electrical systems, plumbing, process gases, controls, and other MEP services collectively establish an organizational structure capable of being planned, constructed, operated, maintained, and evolved as a coherent whole.

That organizational structure is what constitutes a Module.

A Module, therefore, should not be understood as a room type, an equipment package, or a standardized layout. Rather, it is the organizational unit formed around Engineering Capability. Engineering Capability is the object being organized; the Module is the organizational outcome of that process. It is not the starting point of organization, but the stable organizational boundary that gradually emerges as Engineering Capability is continuously sustained throughout the laboratory lifecycle.

This conclusion, however, immediately raises another question.

Engineering Capabilities differ in scale, complexity, and organizational characteristics. Clearly, not every Engineering Capability justifies an independent Module, nor does every Laboratory Cell necessarily evolve into one.

What, then, distinguishes those Engineering Capabilities that truly constitute a Module from those that do not?

3. When Does a Module Truly Exist?

The previous section established that the true object of laboratory organization is **Engineering Capability**, and that a **Module** is the organizational unit formed around it. This naturally raises another question.

If Engineering Capability is the fundamental object of organization, does every Engineering Capability constitute a Module?

The answer is clearly **no**.

A single Engineering Capability can never function as a complete engineering organizational unit on its own. Every laboratory capability depends upon numerous other capabilities to operate. Even the simplest experimental activity requires ventilation, electrical power, utilities, environmental control, monitoring, and supporting infrastructure. A single capability therefore describes what must be achieved, but not an organizational unit capable of sustaining itself.

Engineering Capability is the fundamental object of organization, but it is not yet the fundamental unit of organization.

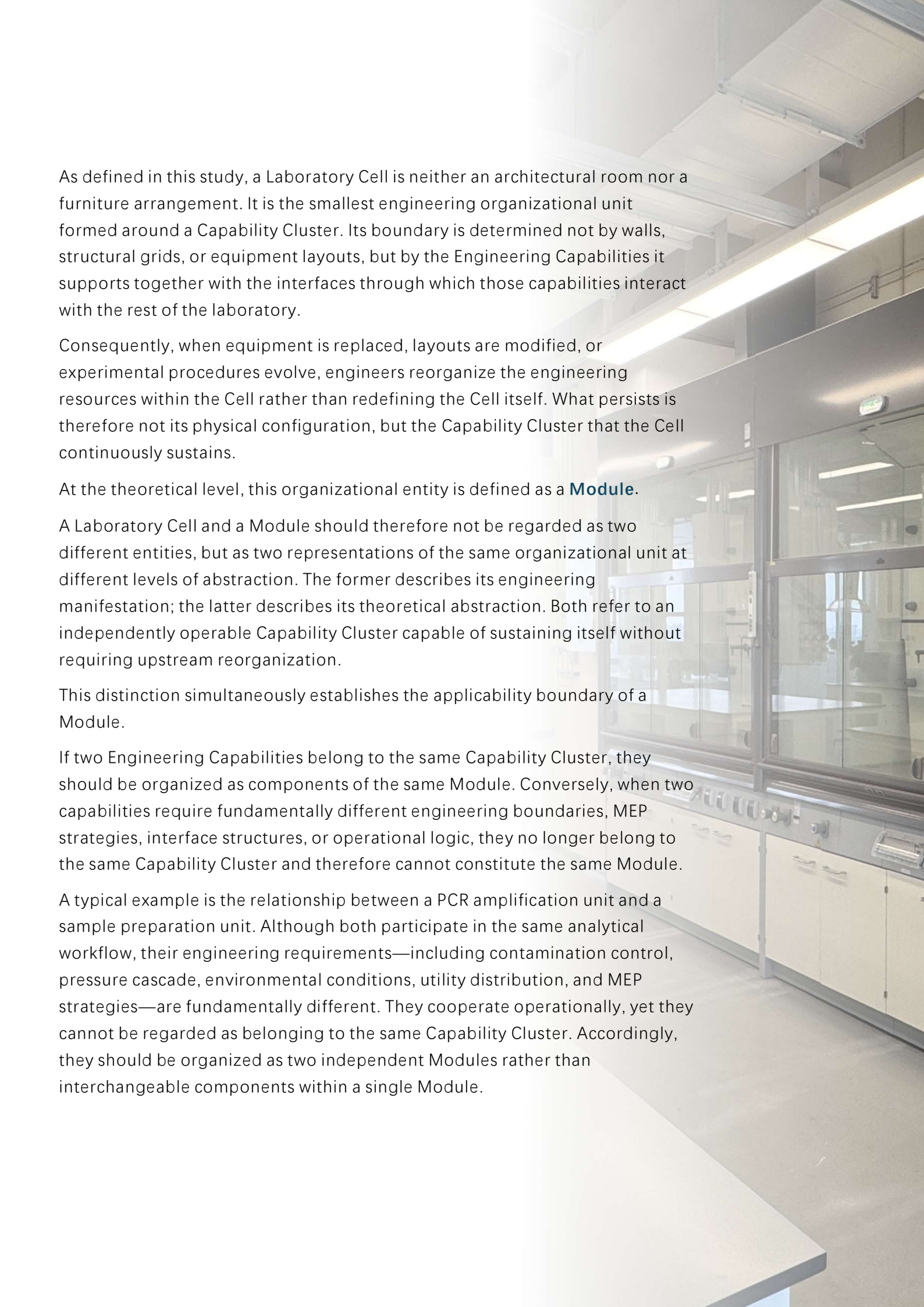
Consequently, what constitutes an organizational unit is not an individual Engineering Capability, but a collection of interdependent capabilities capable of functioning as an integrated whole.

This study defines such an organizational entity as a **Capability Cluster**.

A Capability Cluster is not merely a collection of individual capabilities. Rather, it is an engineering system whose constituent capabilities operate collectively toward a common engineering objective. Internally, the capabilities remain functionally interdependent. Externally, however, the Cluster behaves as a single engineering object that can be planned, constructed, operated, maintained, and modified as one coherent organizational entity.

More importantly, a Capability Cluster can continue operating without requiring the upstream reorganization of the broader laboratory system. Internal modifications may occur continuously, yet those changes remain largely contained within the Cluster itself.

In engineering practice, such a Capability Cluster is typically manifested as a **Laboratory Cell**.

A photograph of a laboratory environment. In the foreground, there is a white countertop. Behind it, a large biosafety cabinet with glass doors and a control panel is visible. The background shows more laboratory equipment and a ceiling with exposed ductwork and lights.

As defined in this study, a Laboratory Cell is neither an architectural room nor a furniture arrangement. It is the smallest engineering organizational unit formed around a Capability Cluster. Its boundary is determined not by walls, structural grids, or equipment layouts, but by the Engineering Capabilities it supports together with the interfaces through which those capabilities interact with the rest of the laboratory.

Consequently, when equipment is replaced, layouts are modified, or experimental procedures evolve, engineers reorganize the engineering resources within the Cell rather than redefining the Cell itself. What persists is therefore not its physical configuration, but the Capability Cluster that the Cell continuously sustains.

At the theoretical level, this organizational entity is defined as a **Module**.

A Laboratory Cell and a Module should therefore not be regarded as two different entities, but as two representations of the same organizational unit at different levels of abstraction. The former describes its engineering manifestation; the latter describes its theoretical abstraction. Both refer to an independently operable Capability Cluster capable of sustaining itself without requiring upstream reorganization.

This distinction simultaneously establishes the applicability boundary of a Module.

If two Engineering Capabilities belong to the same Capability Cluster, they should be organized as components of the same Module. Conversely, when two capabilities require fundamentally different engineering boundaries, MEP strategies, interface structures, or operational logic, they no longer belong to the same Capability Cluster and therefore cannot constitute the same Module.

A typical example is the relationship between a PCR amplification unit and a sample preparation unit. Although both participate in the same analytical workflow, their engineering requirements—including contamination control, pressure cascade, environmental conditions, utility distribution, and MEP strategies—are fundamentally different. They cooperate operationally, yet they cannot be regarded as belonging to the same Capability Cluster. Accordingly, they should be organized as two independent Modules rather than interchangeable components within a single Module.

The boundary of a Module is therefore determined neither by spatial partitions, architectural disciplines, equipment categories, nor laboratory functions. Instead, it is defined by the boundary of the Capability Cluster itself. Only within that boundary can organizational changes remain localized without requiring the reorganization of a larger engineering system.

One final question, however, remains unresolved.

The existence of a Capability Cluster explains when a Module can exist, but not why it can continue to exist. What allows a Module to preserve its organizational identity while the laboratory continues to evolve will be the focus of the next section.



4. What Allows a Module to Remain a Module?

The previous sections established that the true object of laboratory organization is **Engineering Capability**, and that a **Module** is the organizational unit formed around an independently operable **Capability Cluster**. Together, these arguments explain what constitutes a Module and under what conditions it can exist.

One final question, however, remains.

Laboratories are never static. Equipment is continuously replaced, research objectives evolve, workflows are revised, and new engineering requirements emerge throughout the laboratory lifecycle. If a Module were merely the organizational result of a Capability Cluster at a particular moment, then every significant change would require the Capability Cluster itself to be redefined. Under such circumstances, a Module would represent only a temporary organizational arrangement rather than a stable organizational unit.

The persistence of a Module therefore depends not on how it is formed, but on whether it can preserve its organizational identity while continuously adapting to change.

The first requirement for preserving that identity is **Relative Independence**.

Relative Independence does not imply that a Module operates in isolation from the rest of the laboratory. Rather, it means that most organizational changes can be accommodated within the Module itself without requiring the simultaneous reorganization of neighboring Modules. Equipment may be upgraded, furniture reconfigured, local MEP systems optimized, and even experimental procedures revised, yet these adjustments should remain largely confined within the Module's own organizational boundary. Only under such conditions can the Module continue to function as an independent organizational unit.

Relative Independence alone, however, is insufficient.

If every internal modification changes the way a Module interacts with the rest of the system, then the surrounding Modules must also be reorganized, causing change to propagate throughout the laboratory. A Module must therefore also maintain **Stable Interfaces**.



The importance of interfaces lies not simply in connecting Modules, but in isolating change. From the perspective of the surrounding system, a Module should continue to deliver its Engineering Capability through a consistent set of interfaces, regardless of internal modifications. Equipment may change, layouts may evolve, and workflows may be optimized, but these internal adjustments should not continually alter the Module's external relationships. Stable interfaces therefore prevent local evolution from becoming system-wide reorganization.

When Relative Independence and Stable Interfaces coexist, a more fundamental organizational property naturally emerges.

The Module becomes capable of evolving without losing its identity.

Its internal organization may continue to change, yet the Capability Cluster it represents remains organizationally intact. Changes become localized rather than systemic. Capability Clusters no longer require continual redefinition, and organizational evolution can proceed without repeatedly restructuring the laboratory as a whole.

Relative Independence and Stable Interfaces should therefore not be regarded as design preferences introduced for the purpose of modularity. They are the fundamental organizational conditions that allow a Capability Cluster to persist as the same Module throughout successive stages of construction, operation, technological evolution, and organizational adaptation.

Ultimately, a Module is not defined by the permanence of its physical components, but by the continuity of its organizational identity. As long as that identity is preserved, the Module remains the same organizational unit despite continual changes to the means through which its Engineering Capability is delivered.

Only under these conditions can a laboratory continuously adapt to future uncertainty without repeatedly reconstructing its entire organizational system.

Conclusion

The preceding discussion progressively established the theoretical foundation of the Laboratory Module.

First, this chapter demonstrated that the true object of laboratory organization is neither space, equipment, nor workflow, but Engineering Capability—an organizational object capable of persisting across different physical implementations. Building upon this premise, a Module was shown to be an organizational unit formed around Engineering Capability rather than around any particular physical entity.

The discussion then demonstrated that not every Engineering Capability constitutes a Module. Only when multiple interdependent capabilities form an independently operable Capability Cluster, capable of functioning without requiring upstream reorganization, does a Module truly emerge. This distinction simultaneously establishes the applicability boundary of a Module: different Capability Clusters, defined by different engineering objectives, MEP strategies, and interface relationships, cannot be regarded as the same Module.

Finally, this chapter explained why a Module is able to persist throughout continual organizational evolution. The existence of a Capability Cluster establishes the foundation of a Module, while Relative Independence and Stable Interfaces preserve its organizational identity by containing change within its own boundary rather than allowing it to propagate throughout the system. Together, these organizational conditions enable a Module to remain a stable organizational unit despite continual technological and operational evolution.

Based on the foregoing analysis, this study defines a Laboratory Module as follows:

A Laboratory Module is the fundamental organizational unit for organizing Engineering Capabilities. It consists of an independently operable Capability Cluster that maintains its organizational identity while continuously evolving without requiring upstream reorganization.